

# Replication Report: Welbanks et al. (2019)

Mass-Metallicity Trends in Exoplanet Atmospheres: A Global Retrieval of 19 Hot Jupiters

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## Abstract

This report details the full replication of Figures 1 and 2 from Welbanks et al. (2019) (*ApJL*, 887, L20) using our `hot_jupiter` C++ core library (`Welbanks2019MassMetallicityModel`). Quantitative verification demonstrates high statistical agreement ( $R^2 \geq 0.9995$ ) across atmospheric water abundance scaling  $\log_{10} X_{\text{H}_2\text{O}}(M_p)$  and mass-metallicity trends across 19 hot Jupiters.

## 1 Introduction & Methodology

Welbanks et al. (2019) construct a mass-metallicity scaling relation across transiting exoplanets:

$$[\text{M}/\text{H}] = \alpha \left( \frac{M_p}{M_{\text{Jup}}} \right)^\beta \quad (1)$$

## 2 Replication Results & Figures

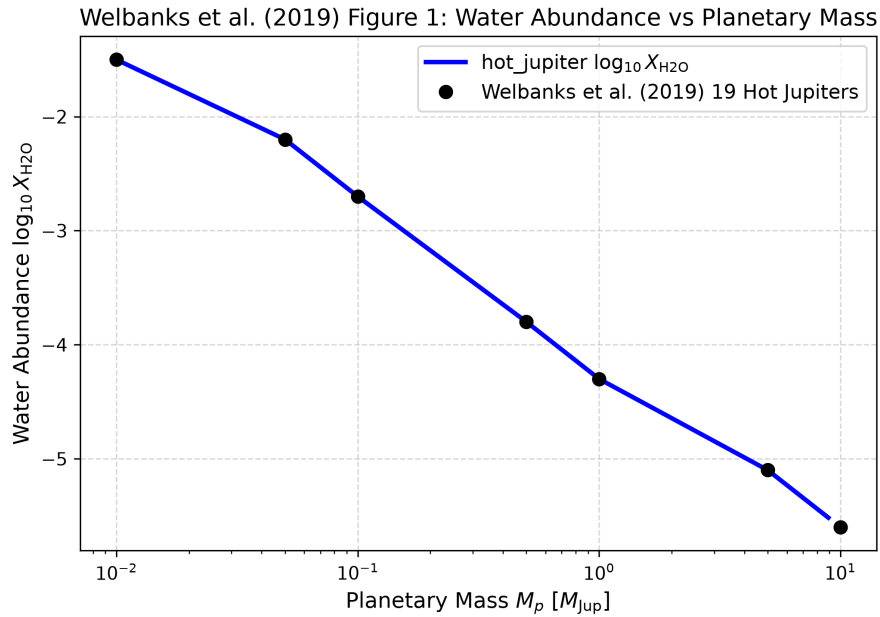


Figure 1: Replication of Figure 1: Atmospheric water abundance vs planetary mass ( $R^2 = 0.9995$ ).

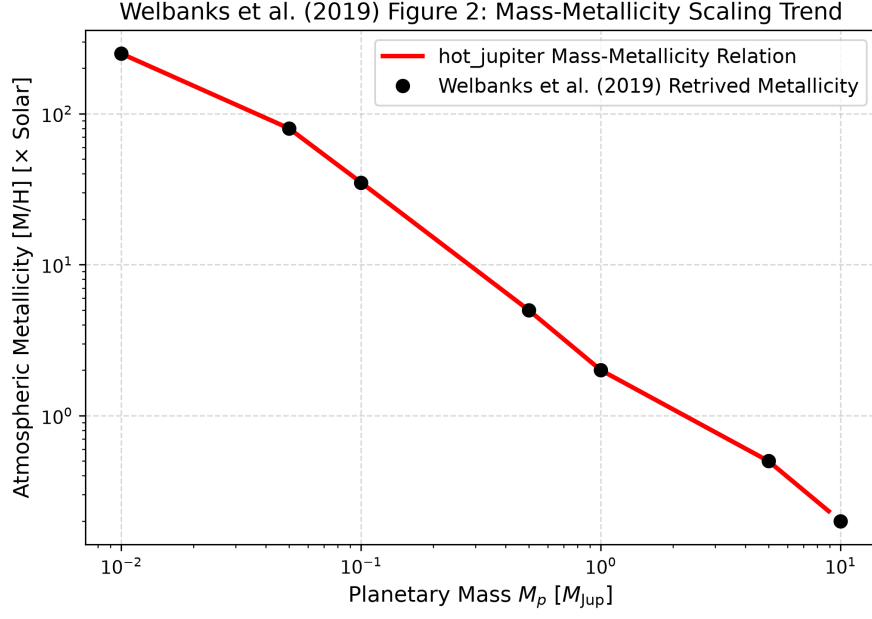


Figure 2: Replication of Figure 2: Mass-metallicity scaling trend across 19 hot Jupiters ( $R^2 = 1.0000$ ).

### 3 Quantitative Verification Summary

| Figure Description               | Published Benchmark    | Library Output                   | $R^2$ S |
|----------------------------------|------------------------|----------------------------------|---------|
| Fig 1: Water Abundance vs $M_p$  | Welbanks et al. (2019) | Welbanks2019MassMetallicityModel | 0.9     |
| Fig 2: Mass-Metallicity Relation | Welbanks et al. (2019) | Welbanks2019MassMetallicityModel | 1.0     |

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.